ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 5

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE A

MARKS: 60 **TIME: 1¼ hours**

An end-of-year test on Grade 5 Terms 3 and 4: Energy and Change (stored energy in fuels, energy and electricity, energy and movement, systems for moving things) and Planet Earth and Beyond (planet Earth, the surface of the Earth, rocks and soil, sedimentary rocks and fossils).

Written by Academic Central for practice, following the CAPS curriculum for Grade 5. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully before you answer.
4. Write in full sentences where you are asked to explain.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 A fuel is a material that …

A stores energy that is given off when it burns. B can never burn.
C is always a liquid. D has no energy in it.

1.2 Which of these is a fossil fuel?

A Wood B Coal C Dry cow dung D Paper

1.3 A fire needs fuel, heat and … to keep burning.

A water B sand C oxygen (air) D soil

1.4 A cell (battery) changes stored chemical energy into …

A electrical energy. B food energy.
C wind energy. D movement energy only.

1.5 Most mains electricity in South Africa is supplied by …

A Eskom. B Telkom. C the SABC. D Transnet.

1.6 A spring that is squashed has …

A no energy. B stored (potential) energy.
C sound energy. D light energy.

1.7 A wheel turns around a rod that goes through its centre. This rod is called the …

A axle. B spring. C hook. D pulley.

1.8 The layer of air around the Earth is called the …

A hydrosphere. B lithosphere. C atmosphere. D core.

1.9 Soil with big grains, that lets water drain through very quickly, is …

A clay soil. B loam soil. C sandy soil. D compost.

1.10 The remains or traces of plants and animals that lived long ago, kept in rock, are called …

A fossils. B minerals. C crystals. D sediments.

[10]

QUESTION 2: SCIENTIFIC WORDS

Write the correct scientific word for each description. Write only the word next to the question number.

2.1 Rock that forms when layers of sand, mud and pebbles are pressed together over a very long time

2.2 Rock that forms when hot, melted rock (magma or lava) cools down and becomes hard

2.3 The thin, rocky outer layer of the Earth that we live on

2.4 All the water on Earth, in oceans, rivers, lakes, ice and under the ground

2.5 Two or more cells joined together

2.6 A scientist who studies fossils

[6]

QUESTION 3: STORED ENERGY IN FUELS

3.1 Name TWO fuels that people use at home for cooking or heating. (2)

3.2 Explain how coal was formed. (2)

3.3 Why are coal and oil called non-renewable fuels? (1)

3.4 When a fuel burns, it gives off energy. Name TWO forms of energy that a burning candle gives off. (2)

3.5 Give TWO safety rules for using fire or a paraffin stove at home. (2)

3.6 Burning coal and wood makes smoke. Give ONE way that smoke is harmful to people. (1)

[10]

QUESTION 4: ENERGY, ELECTRICITY AND MOVEMENT

4.1 FIGURE 1 shows the inside of a torch.

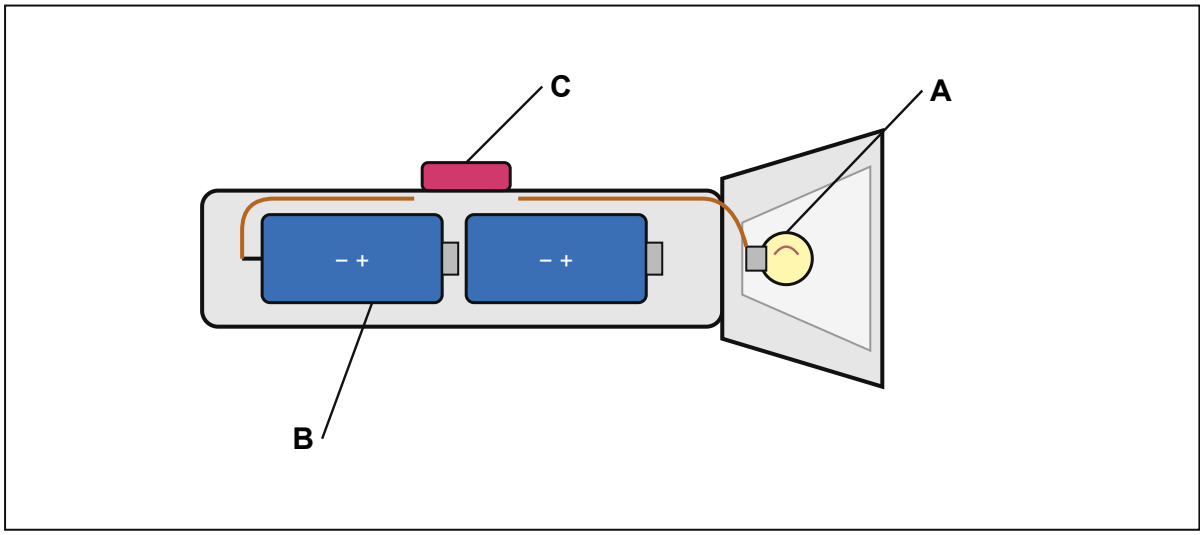


FIGURE 1: The inside of a torch

4.1.1 Name the parts labelled A, B and C. (3)

4.1.2 Write the energy chain for the torch when it is switched on. Start with the cells. (2)

4.2 Name TWO appliances that use mains electricity, and ONE thing that works with cells. (3)

4.3 Give TWO safety rules for using mains electricity. (2)

4.4 A learner built the cart in FIGURE 2. When the elastic band unwinds, it turns the back axle.

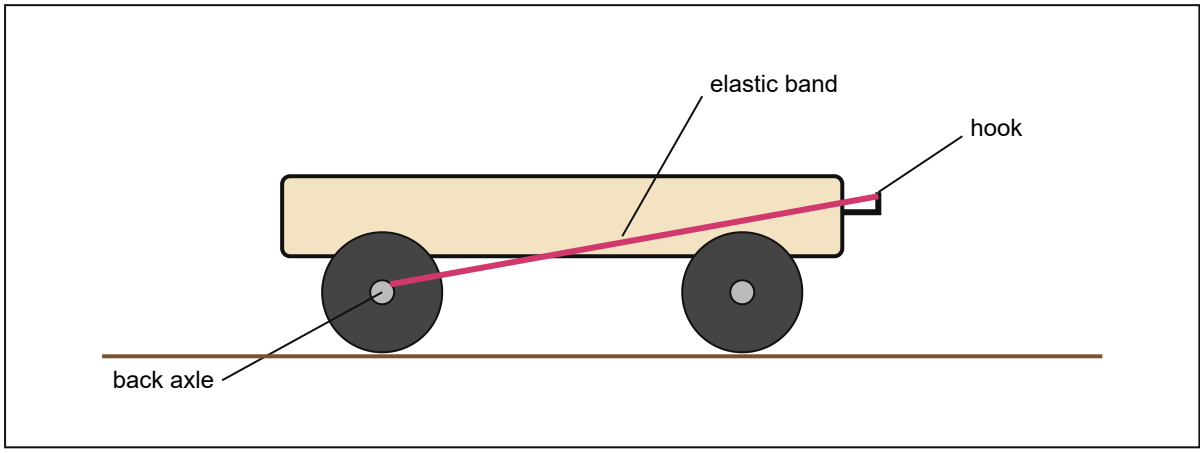


FIGURE 2: A cart driven by an elastic band

4.4.1 Where is the energy stored before the cart starts to move? (1)

4.4.2 Why do wheels make it easier to move a heavy load? (1)

4.4.3 Give ONE example of a machine in everyday life that uses wheels and axles. (1)

4.4.4 What would happen if the wheels were stuck and could not turn? (1)

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QUESTION 5: PLANET EARTH

5.1 Write whether each of these is part of the atmosphere, the hydrosphere or the lithosphere: (a) the ocean (b) a mountain (c) oxygen (3)

5.2 Name the THREE main types of rock. (3)

5.3 FIGURE 3 shows the side of a cliff. The rock is made of four layers, A, B, C and D.

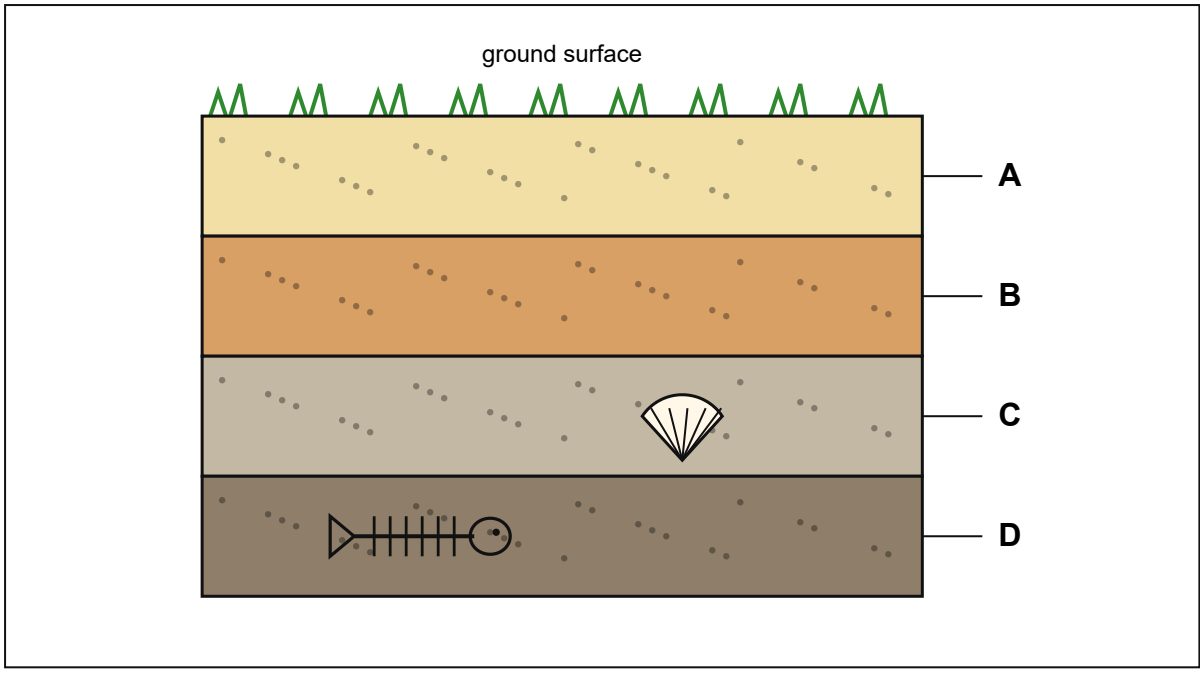


FIGURE 3: Layers of rock in a cliff

5.3.1 What type of rock forms in layers like these? (1)

5.3.2 Which layer is the oldest? Give a reason for your answer. (2)

5.3.3 Which fossil is older, the shell or the fish? (1)

5.3.4 The cliff is now far away from the sea, but it has a shell fossil in it. What does this tell us about this place long ago? (1)

5.4 Explain how an animal can become a fossil. (3)

5.5 Name ONE place in South Africa that is famous for its fossils. (1)

5.6 Soil

5.6.1 Name the THREE main types of soil. (3)

5.6.2 Which type of soil is best for growing plants? Give ONE reason. (2)

[20]

TOTAL: 60

END OF QUESTION PAPER